End-to-end capstone for one endpoint

Session M6-02 | 3 hours

One slice

Implement and review:

```
POST /api/bookmarks
```

Everything else is deferred.

The result is a handoff

- implementation-ready issue;
- implemented patch;
- executable test results;
- review decision;
- deferred scope and next action.

The scope lives in the issue

Implement only `POST /api/bookmarks`, use synthetic data, and keep the existing dependencies.

Stop if the work requires another endpoint, live data, deployment, tool configuration, or a new dependency.

Session M5-02 is an input

The accepted specification must separate:

Behavior	Implementation constraint
Observable request and response rules	Node.js, Express, in-memory storage
Validation and duplicate behavior	Existing dependencies only
Acceptance scenarios	Exact files and test command

Issue trace

Every acceptance criterion maps to:

```
specification ID -> planned file -> executable test
```

If the issue adds behavior missing from the specification, revise the issue before implementation.

Scope guard

In scope:

- create one bookmark;
- validate the accepted fields;
- reject a duplicate URL;
- return the documented envelope and status.

Out of scope: authentication, persistence, other endpoints, deployment, and new packages.

When work is blocked

Use GitHub Copilot for the implementation.

Record the blocker and a request-changes or pause decision in the handoff.
Resume the work before claiming completion.

Run the tests

```
npm test
```

The suite checks success, defaults, field validation, and duplicate URLs.

Review and handover

Trace the change back to the specification. Review the diff, test output, dependency change, and scope decisions in the issue. Record any deviation in the handoff.

Record one decision: approve, request changes, or pause.

Lab time plan

Phase	Time
Accept the specification and scope	25 min
Write the issue and establish the baseline	25 min
Implement the accepted slice	35 min
Test, review, and hand off	35 min